Session 1: Getting Started With Red Hat Linux

- Understand the Linux distro and RHEL.
- Manage Files, Users, Groups, and Permissions.
- Install and manage software (yum/dnf).

Session 2: Red Hat Linux Essentials

- Manage and control Background Services
- Configure and troubleshoot Linux Networking.
- Understand advanced File Permissions.

Session 3: Shell Scripting

- Shell Scripting Basics
- Add logic with Control Flow
- Basics of User & Group Management
- **Project: Build a Complete Automation project for user and group management with logs.**

Session 4: Shell Scripting: Filesystems & Performance Monitoring

- Automate Filesystem tasks
- Create a System Metrics Script with Alerts
- **Project: Make a project that combines filesystem automation and system performance reporting.**

Session 5: AWS Global Infrastructure & EC2

- Understand the Global AWS Structure.
- Introduction of EC2
- Launch your first EC2 instance.

Session 6: Instance Configuration & Connectivity

- Understand AWS Connectivity.
- Advanced EC2 options.
- Prepare AI-ready compute (Golden AMIs and AWS Systems Manager).
- Build and test a secure baseline EC2 setup.

Session 7: Advanced AWS Services

- Understand AWS Network Security Layers.
- Elastic Load Balancing and Auto Scaling.
- Amazon S3 fundamentals
- Store and serve datasets or ML models

Session 8: CloudFormation, CloudWatch & Serverless

- Learn Infrastructure as Code (IaC)
- Overview of RDS and IAM.
- Understand SNS and SQS.
- AWS Lambda and API Gateway.
- **AI Project: Implement AI-ready observability patterns using CloudFormation, CloudWatch, and SNS/SQS.**

Session 9: Mastering Git

- Distributed and Centralized Version Control.
- Git Three-Stage Architecture
- Learn to manage AI assets

Session 10: Advanced Git & GitHub

- Initialize and push your infra & automation repos
- GitHub auth, PR workflows, and code review etiquette.
- Advanced Git commands
- MLOps Collaboration Workflows
- GitHub Copilot for automating commit messages, PR reviews, and generating initial CI/CD workflow YAMLs.

Session 11: Networking Fundamentals

- Learn IP addressing, subnetting, and CIDR
- Understand system communication.
- DNS fundamentals.
- Design a VPC subnet plan and visualize routing paths

Session 12: Conceptual Concepts of Dockers

- Understand Virtualization vs. Containerization.
- Docker Architecture.
- Docker images and lifecycle

- **Project: Containerize an inference service using FastAPI or TorchServe**

Session 13: Docker: Images, Networking, Volumes & CI/CD

- Understand Docker Networking Modes
- Explore container security and compliance
- Implement CI/CD pipelines
- **Project: Document image usage and operational procedures.**

Session 14: SonarQube, Quality Gates, and Profiles

- Code Quality in Operations.
- Set up SonarQube
- Enforce Quality Gates

Session 15: CI/CD Workflow with Jenkins

- Understand CI/CD fundamentals.
- Explore Jenkinsfile syntax
- Understanding Artifact Versioning and Management
- Apply CI/CD to an ML workflow
- Use GitHub Copilot to scaffold Jenkinsfiles.

Session 16: Advanced Jenkins & GitLab Integration

- Understand Jenkins agents and Executors
- Controller-Agent models and Dynamic Agent
- Identify key Jenkins plugins(Docker, Git, Maven,)
- GitLab CI/CD concepts
- Implement an advanced multi-stage ML pipeline.
- Explore Jenkins AI plugins
- **Project: Set up a GitHub Actions self-hosted runner and trigger it through Jenkins for hybrid automation**

Session 17: Kubernetes Introduction

- Understand Container Orchestration
- Learn Kubernetes Architecture components
- Use kubectl and minikube.
- Deploy your first pod and service
- Overview of Azure DevOps

Session 18: Kubernetes Basics

- Kubernetes workloads with Deployments
- Understand labels, selectors, rollouts, rollbacks, and probes
- Explore multi-container pod patterns.
- Basics of KServe and Seldon
- Perform a blue/green rollout.

Session 19: Deploying Applications on Kubernetes

- Understand EKS
- Minikube vs. EKS
- Configure Node Groups and IRSA
- Build a Kubernetes Resilience with HPA & Rollout Strategies.
- **Project: Deploy a complete microservice application to your EKS cluster with health checks and persistent volumes**

Session 20: Advanced Kubernetes Topics

- RBAC and Service Accounts control permissions
- Understand Ingress concepts
- Package and deploy complex applications (Helm charts)
- Fundamentals of Service Mesh (Istio).
- **Project: Deploy a TLS-secured EKS application using Nginx Ingress and Cert-Manager, and analyze monolith vs. microservices tradeoffs**

Session 21: Getting Started with Ansible

- Ansible for Configuration Management
- Ansible Architecture
- Ansible Automation with Roles, Collections, Vars & Facts

Session 22: Automation with Ansible Playbooks

- Organize playbooks with tags.
- Jinja2 templating
- Secure secrets with Ansible Vault
- **Project: Automate a multi-node Kubernetes cluster setup using repeatable, idempotent playbooks.**

Session 23: Terraform: Introduction & State

- Understand IaC principles
- Terraform Basics
- Understanding provider Authentication methods.
- Design and provision a VPC architecture with subnets, route tables, and gateways.

Session 24: Terraform Advanced + Grafana & Prometheus

- Advanced Terraform concepts
- Understand Networking with CIDR /16
- Configure drift alerts and FinOps monitoring
- Set up and integrate Grafana, Prometheus, and MySQL
- **AI Project: Build observability dashboards for ML models - tracking latency, accuracy, and error rates.**